1.1 DISTANCE, DISPLACEMENT, SPEED AND VELOCITY

LEARNING GOALS

Students will:

- Review the definitions of distance displacement position, speed and velocity.
- Perform vector and/or scalar calculations on the above quantities.

REVIEW OF MOTION CONCEPTS

THE FLYING BALL

Vince throws a ball straight up. It rises to its maximum height and then Simeon catches it 1.2 m above Vince's hand on its way back down.

List all the known information about the ball's motion. If the info is not given, leave it blank

d	distance	$\vec{a}$	acceleration $\vec{a}_x = 0 m/s^2$ $\vec{a}_y = -9.8 m/s^2$
$\vec{d}$	positian At catch: +1.2m [up]	$\vec{v}_i$	initial velocity
$\Delta\vec{d}$	displacement $\Delta\vec{d}$ = +1.2m [up]	$\vec{v}_f$	final velocity
$\vec{v}_{av}$	overage velocity	$\Delta\vec{v}$	change in velocitg

THE DIFFERENCE BETWEEN $\vec{v}_{av}$ AND $\Delta\vec{v}$

A Porsche Carrera accelerates from 80.0 km/h to 140. km/h in 1.8 s and covers 55 m.

Find $\vec{v}_{av}$ and $\Delta\vec{v}$.

$\vec{V}_i = 80 km/h$ $\quad \Delta\vec{d}_x = 55m$ $\quad V_f = 140 km/h$

$\vec{d}_i = 0m$ $\quad \Delta t = 1.8s$ $\quad \vec{d}_f = 55m$

- ⟷ +

① $\vec{V}_{av} = ?$

$$\vec{V}_{av} = \frac{\Delta\vec{d}}{\Delta t}$$

$$= \frac{55m}{1.8s}$$

$$= 31 m/s \text{ [right]}$$

② $\Delta\vec{V} = ?$

$$\Delta\vec{v} = \vec{V}_f - \vec{V}_i$$

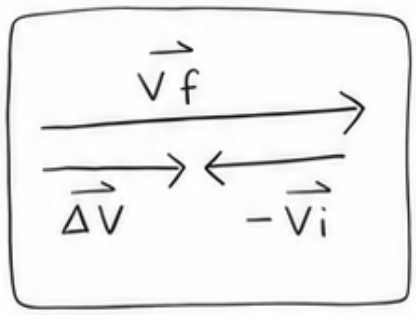
$$\Delta\vec{v} = 140 km/h - 80 km/h$$

$$\Delta\vec{v} = 60 km/h$$

$$\Delta\vec{v} = 60 km/h \times \frac{1000 m}{1 km} \div \frac{3600 s}{1 h}$$

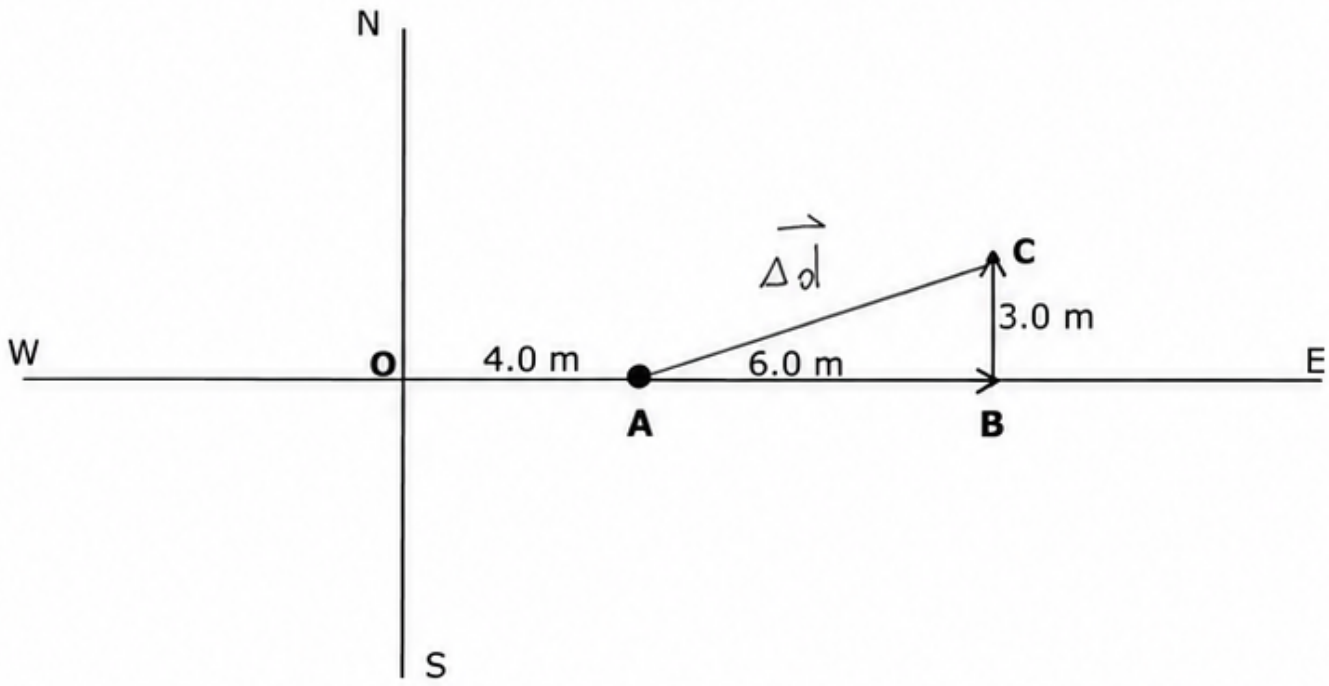
$$\Delta\vec{v} = 60 \cancel{km/h} \times \frac{1/3.6 m/s}{1 \cancel{km/h}}$$

$$\Delta\vec{v} = 17 m/s \text{ [right]}$$



∴ Avg velocity is 31 m/s [right] and the change in velocity is 17 m/s [right]

POSITION, DISPLACEMENT AND DISTANCE



In the above diagram, use the origin **O** as the reference point. A student starts at **A**, walks to **B** in 6.0 s then walks to **C** in another 6.0 s.

POSITION

1. State the position of the student at (Answers at bottom of next page)

A 4.0m [E]

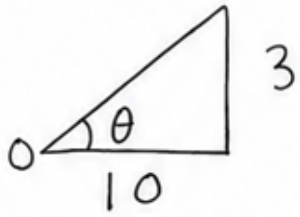
B 10.0m [E]

C (This one needs to be calculated)

10.m [E17°N]

$C' = \sqrt{10^2 + 3^2}$

$C = 10.m$



$\theta = \tan^{-1}\left(\frac{3}{10}\right)$ $\theta = 17°$ North of east

DISTANCE

2. What is the distance walked as the students walks from

A to **B** 6.0 m

B to **C** 3.0 m

A to **C** 9.0m

DISPLACEMENT

3. What is the displacement as the students walks from

A to **B** $\vec{\Delta d}_{AB} = \vec{d}_B - \vec{d}_A$. $\vec{\Delta d}_{AB} = 10\,m[E] - 4.0m[E]$ $\vec{\Delta d}_{AB} = 6.0\,m\,[E]$

B to **C** 3.0m [N]

A to **C** 6.7 m [E27°N]

The displacement from **A** to **C** is the **total displacement.** You found it by adding the individual displacements from **A** to **B** and from **B** to **C**. Therefore total displacement can be found by

$$\overrightarrow{\Delta d_T} = \overrightarrow{\Delta d_1} + \overrightarrow{\Delta d_2}$$

There is another way to find total displacement. **Displacement is defined to be the change in position**. This means that displacement is given by final position minus initial position

$$\overrightarrow{\Delta d_T} = \overrightarrow{d_2} - \overrightarrow{d_1}$$

4. Try it. Take your final position at C (found above) and your initial position at **A** and do a vector subtraction. You should get the same total displacement you got above from **A** to **C**.

$\vec{\Delta d}_{tot} = \vec{d}_C - \vec{d}_A$

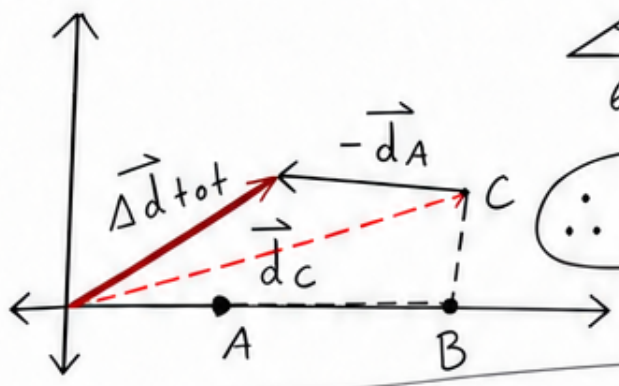
$\vec{\Delta d}_{tot} = (10.0, 3.0)m - (4.0, 0)m$

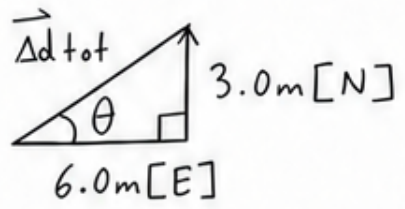
$\vec{\Delta d}_{tot} = (10.0 - 4.0, 3.0 - 0)m$

$\vec{\Delta d}_{tot} = (6.0, 3.0)m$

$\vec{d}_C = (10.0, 3.0)m$ → coordinates for positions

$\vec{d}_A = (4.0, 0)m$





$\Delta d_{tot} = \sqrt{3.0^2 + 6.0^2}$

$\Delta d_{tot} = 6.7m$

$\theta = \tan^{-1}\left(\frac{3}{6}\right)$

$\therefore \vec{\Delta d}_{tot} = 6.7m[E27°N]$ $\theta = 27°$ north of east

Algebraic method point-to-point

$\vec{\Delta d}_{tot} = \vec{d}_C - \vec{d}_A$

$= \vec{O}_C - \vec{O}_A$

$= \vec{O}_C + AO$

$= AO + O_C$

$= \boxed{\vec{AC}}$

$\vec{\Delta d}_{tot} = 6.7m[E27°N]$ $\vec{\Delta d}_{AC} = 6.7[E27°N]$

$\therefore \vec{\Delta d}_{tot} = \vec{\Delta d}_{AC}$

AVERAGE SPEED

5. Definition: The rate at which distance is covered. Write the equation for this:

$$V_{avg} = \frac{\Delta d}{\Delta t}$$

6. Find the average speed of the student when walking from (Answers at bottom)

A to **B** $\frac{6.0\,m}{6\,s} = 1\,m/s$

B to **C** $\frac{3\,m}{6\,3} = 0.50\,m/s$

A to **C** $\frac{9\,m}{12\,s} = 0.75\,m/s$

AVERAGE VELOCITY

7. Definition: The rate of change of **position**. Write the equation for this:

$$\vec{V}_{av} = \frac{\vec{\Delta} d}{\Delta t}$$

8. Find the average velocity of the student when walking from

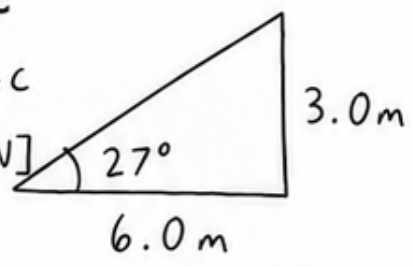
A to **B** $= 6\,m\,[E] / 6\,s \quad = 1.0\,m/s\,[E]$

B to **C** $= 3\,m\,[N] / 6\,s \quad = 0.50\,m/s\,[N]$

A to **C**

$\vec{\Delta} d_{AC}$

$= 6.7\,m\,[E27°N]$



$= \frac{6.7\,m\,[E27°N]}{12\,s}$

$= 0.5590\,m/s\,[E\,27°N]$

$\vec{V}_{avAC} = 0.56\,m/s\,[E\,27°N]$

9. What do you think it means when a definition uses the word "rate"?

rates measure the change in quantity per unit of time.

PRACTICE

Text p 10 #1,3 – 5, p 16 #1, p 26 #1, p 28 #1 - 3

Answers for Notes

1. **A** 4.0 m[E] **B** 10.0 m[E] **C** 10.4 m[E17°N]
2. 6.0 m 3.0 m, 9.0 m
3. 6.0 m [E] 3.0 m [N] 6.7 m [E27°N]
6. 1.0 m/s 0.50 m/s, 0.75 m/s
8. 1.0 m/s [E] 0.50 m/s [N] 0.56 m/s [E27°N]